Results on Mistral-7B-v0.2. The top part of Fig. 5 shows that IFT and ICL on SkillMix (violet and green curves respectively) perform almost identically in 1st-turn score until 50 examples are used. Beyond this point, increasing the number of training examples consistently improves the performance of IFT. The strong performance of IFT in the low-data regime is particularly surprising, as one might expect overfitting when training on very few examples (as few as 3) over several epochs, yet this does not occur. Finally, ICL and IFT show nearly opposite trends for the 2nd-turn score: increasing the dataset size benefits IFT (almost reaching the performance of the instruct model) but detrimental to ICL. This behavior suggests that, while IFT is less flexible due to the model weights being updated, it generalizes better to tasks different from those used for alignment (recall that the instructions in SkillMix are single-turn only).

Results on Llama-3.1-7B. As shown in the bottom part of Fig. 5, ICL outperforms IFT when using between 3 and 30 demonstrations, though IFT remains effective even with these few examples on this base model. However, we note that IFT matches or exceeds the best performance of ICL (obtained with 20-30 examples) only when using 2k or 4k examples (the entire SkillMix), which represents two orders of magnitude more data. This result confirms that ICL with high-quality data is a viable alternative to IFT when only a limited number of demonstrations are available. Finally, the observations for the 2nd-turn scores are consistent with those for Mistral-7B-v0.2, with IFT consistently outperforming ICL.

Effect of data quality. Finally, we test the effect of using lower-quality instructions compared to SkillMix. In this experiment, we repeat the ICL vs. IFT comparison using random demonstrations from Evol-Instruct-70k (Xu et al., 2024), and show the results in Fig. 5. With Evol-Instruct data, both ICL (red curve) and IFT (yellow curve) perform significantly worse than their counterparts using SkillMix. This trend is consistent across the number of examples, base models, and for both single-turn and multi-turn instructions. Interestingly, the performance gap between IFT on SkillMix and Evol-Instruct is significantly smaller on Llama-3.1-8B than on Mistral-7B-v0.2, suggesting that better pre-trained models may partially compensate for lower-quality fine-tuning data. Finally, we notice that ICL with 3 examples from Evol-Instruct gets worse 1st-turn scores than URIAL (with also 3 examples), whereas ICL with 3 examples from SkillMix matches (on Mistral-7B-v0.2) or outperforms (on Llama-3.1-8B) URIAL (see also discussion in Sec. 3.1). These observations further confirm the importance of instruction quality for alignment, whether in the context of in-context learning or instruction fine-tuning.

5 Conclusions

In this work, we have first illustrated that ICL via URIAL is a good baseline for instruction-following alignment, but with a few limitations: it typically performs slightly worse than IFT on single-turn conversations and does not generalize well to multi-round ones. Then, we have uncovered the key components for IC alignment: e.g. the decoding parameters alone can, surprisingly, make base models reasonably good at instruction following. Adding in-context high-quality demonstrations improves performance beyond what previously suggested (Lin et al., 2024), but not enough to reach LLMs aligned with sophisticated methods (e.g., RLHF). We conjecture that via ICL, the LLM can learn to infer the response style, but the overall learning signal is not sufficiently strong to benefit from a large amount of examples, despite the long context windows of recent LLMs.

Moreover, to the best of our knowledge, we have provided the first systematic comparison of ICL and IFT for instruction following when using the same (small) number of demonstrations. Surprisingly, the two approaches are roughly equivalent in terms of single-turn conversation performance. However, models trained via IFT, unlike ICL, can generalize to multi-round conversations, which are out-of-distribution compared to the examples used for alignment. Overall, our work provides a deeper and more complete understanding of how in-context alignment works, as well as of its potential and limitations, in particular in comparison to fine-tuning. This comprehension might be the basis for future work aimed at using ICL to efficiently customize LLMs without the need of fine-tuning, as well as exploring the fundamental differences between base and aligned models.

Limitations. We perform greedy search with limited computational resources, e.g. the set of candidate examples is relatively small. We expect that more effective ICL prompts could be identified with additional resources. Moreover, our focus is on instruction-following tasks, but other types of alignment, such as safety-oriented tasks, may also be of interest.

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